# ICMA 216 Calculus IIIA: Homework 5

Due on Monday, February 9, 2026

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# Problem 1 (11.7 Quadric Surfaces)

Sketch the following surfaces.

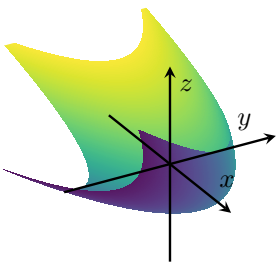
(a) $-9z^2 - 5y^2 - x + 4 = 0$

(b) $4y^2 + 9z^2 - x^2 = 0$

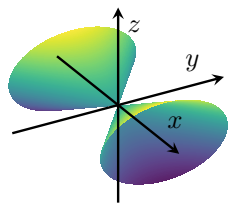
(c) $1 - 3z^2 - y^2 - 4x^2 = 0$

## Solution

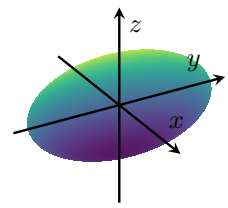
(a) $x = 4 - 5y^2 - 9z^2$




(b) $x^2 = 4y^2 + 9z^2$




(c) $4x^2 + y^2 + 3z^2 = 1$




*Remark.* hand written plots for this problem were attached by submission.

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# Problem 2 (12.1 Vector Functions and Space Curves)

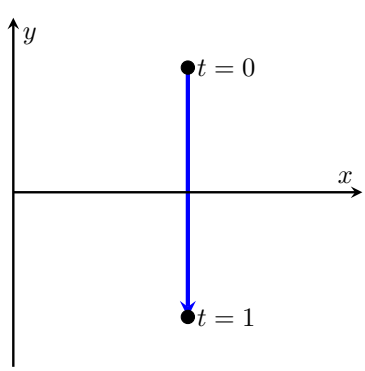
Sketch the following curves.

(a) $r = (1-t)(i+j) + t(i-j); 0 \le t \le 1$

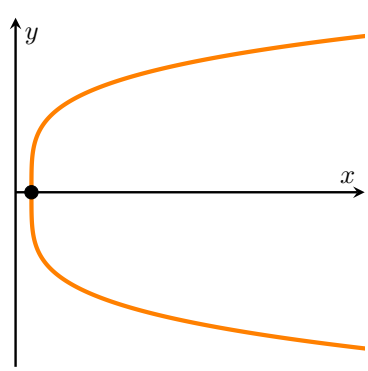
(b) $r(t) = \langle t^4 + 1, t \rangle$.

## Solution

(a) $r(t) = \langle 1, 1 - 2t \rangle$




(b) $x = y^4 + 1$

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# Problem 3 (12.2 Calculus of Vector Functions)

Given $r(t) = (1+t)i + t^2 j$. Find the unit tangent vector at the point where $t = 1$.

## Solution

we could find the velocity vector by

$$r'(t) = \frac{d}{dt}[(1+t)i + t^2 j] = i + 2tj$$

at $t = 1$ gives

$$r'(1) = i + 2j$$

which we could calculate it's magnitude as

$$\|r'(t)\| = \sqrt{1^2 + 2^2} = \sqrt{5}.$$

and therefore the unit tangent vector at $t = 1$ is

$$\frac{r'(1)}{\|r'(1)\|} = \boxed{\frac{1}{\sqrt{5}}(i + 2j)}$$

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# Problem 4 (12.2 Calculus of Vector Functions)

Evaluate the integral

$$\int (e^t i + 2tj + \ln t k)\, dt.$$

## Solution

we just integrate the vector componentwise as

- $i$-component
$$\int e^t\, dt = e^t + C_1$$

- $j$-component
$$\int 2t\, dt = t^2 + C_2.$$

- $k$-component (integrate by parts)
$$\int \ln t\, dt = t \ln t - t + C_3.$$

merge the three components, combine their constants into a single constant vector C gives

$$\int (e^t i + 2tj + \ln t k)\, dt = \boxed{e^t i + t^2 j + (t \ln t - t)k + \mathbf{C}}$$

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# Problem 5 (12.2 Calculus of Vector Functions)

Find $r(t)$ if $r'(t) = t^2 i + 4t^3 j - t^2 k$ and $r(0) = k$.

## Solution

integrating $r^{'(t)}$ componentwise, just like the previous problem as

$$r(t) = \left(\int t^2\,dt\right) i + \left(\int 4t^3\,dt\right) j - \left(\int t^2\,dt\right) k + \mathbf{C} = \frac{t^3}{3} i + t^4 j - \frac{t^3}{3} k + \mathbf{C},$$

where $\mathbf{C}$ is a constant vector, then apply the initial condition $r(0) = k$

$$r(0) = 0i + 0j + 0k + \mathbf{C} = k.$$

therefore, $\mathbf{C} = k$, and hence

$$\boxed{r(t) = \frac{t^3}{3} i + t^4 j + \left(1 - \frac{t^3}{3}\right) k}$$

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